PROJECT HANDOFF

Claude of Duty

A grounded first-person tactical shooter: physical weapon weight, real bore-axis ballistics, picture-in-picture optics, sound that travels at the speed of sound, and enemies that fight as fire teams.

Date	16 October 2026
Engine	Unity 6000.6.4f1 · HDRP 17.6.0 · Input System 1.20.0 · AI Navigation 2.0.14
Repository	github.com/LUK3N4Z/Claude-of-duty · branch `main` · head `1b05b0a`
Status	All 10 milestones complete · 89 PlayMode tests (84 pass, 5 opt-in skipped, 0 fail)
Build	Windows 64-bit player builds and runs (boots into the Armory hub)

Everything in the game — geometry, weapons, parts, grenades, soldiers, materials, sounds, voices, scenes and NavMeshes — is generated by editor tools. Art is primitive-based and audio is synthesized: the systems are complete, the presentation is placeholder.

Contents

1 Status at a glance
2 Getting started
3 Controls
4 Movement and the player's body
5 Weapon handling, aiming and recoil
6 Weapon mechanics and weapons
7 Gunsmith and attachments
8 Ballistics, armour and damage
9 Sound
10 Graphics and rendering
11 Throwables and equipment
12 Enemy AI
13 Game modes
14 Armory, profiles and settings
15 Maps and scenes
16 Architecture
17 Content pipeline and tests
18 Known issues and workarounds
19 Recommended next steps

1 Status at a glance

Area	What exists
Scenes	Armory hub (workshop + indoor range), Locomotion Test Range, Tactical Raid, Search & Destroy, Hardpoint
Weapons	Modular 5.56 carbine; .300 Win Mag bolt-action marksman rifle
Attachments	14 parts across 7 slots, live-swappable, with real mass and balance effects
Ammunition	5.56×45, 7.62×51, 9×19, .300 Win Mag, grenade fragments — all with G1/G7 drag
Equipment	M67 frag, M84 stun, M18 smoke, C4 demolition charge
AI	Soldiers (friendly and hostile) fighting as squads, with CQB room clearing and three voices
Game modes	Tactical Raid / Extraction, Search & Destroy, Hardpoint
Persistence	JSON profile with weapon blueprints, loadout and settings
Tests	89 PlayMode tests (84 pass, 5 opt-in render captures skipped) plus visual captures on demand

Milestones (all complete)

#	Milestone	Commit
M1	Project init, assemblies, InputReader, EventBus, pooling, setup wizard	`80d1fba`
M2	Locomotion biomechanics, stances, leaning, mounting, breathing	`38dafd5`
M3	Springs, weapon inertia, barrel collision, procedural ADS, PiP optics, recoil	`7482bb4`
M4	Weapon/caliber data, model rig and sockets, chamber tracking, staged reloads	`ce375b7`
M5	Ballistics sim, penetration/ricochet, shell casings, health and armour	`bb38a9b`
M6	Acoustic propagation, muzzle VFX, diegetic HUD	`2d5db22`
M7	Gunsmith and attachment physics	`93c506a`
M8	Tactical AI and squad CQB	`65a13f5`
M9	Game modes, match director, tactical equipment	`8367d25`
M10	3D gunsmith workbench, loadout persistence, HDRP settings menu	`cc73f52`
Fix	Editor workaround for an HDRP 17.6 volumetric fog bug	`1b05b0a`

Every milestone was verified with batch builds, the full PlayMode suite and render captures before it was committed.

2 Getting started

Play the built game

Run `Desktop\Claude of duty Build\ClaudeOfDuty.exe`. It opens fullscreen in the Armory hub. Escape opens the options menu; there is no Quit button yet, so exit with **Alt+F4**. The build only contains the hub as a playable entry point — see Known issues.

Open the project

- Unity Hub → Add → project from disk → the repository folder, editor **6000.6.4f1**.
- The builders below are already applied and committed; they are idempotent, so rerun any of them after a settings reset.
- Active Input Handling is *Input System Package (New)*.
- To play in the editor: open a scene under `Assets/_Project/Scenes`, press Play, then **click inside the Game view** so it takes keyboard and mouse focus.

Editor builders (`Tools/FPS Builder`, run in order)

#	Menu	Creates
1	Initialize Project Layers & Tags	Linear colour space, HDRP asset, layers, collision matrix
2	Create Core Assets	`Data/Core/InputReader.asset` wired to the input actions
3	Create Test Weapons	Calibers, carbine and marksman prefabs, data, recoil profiles, magazines, all gunsmith parts
4	Create Player Prefab	Rig, viewmodel, IK arms, loadout, equipment, profile manager, settings menu
5	Build Test Range	Locomotion Test Range, AI soldier prefabs, grenades, voices, Squad Alpha, CQB shoot house, NavMesh
6	Build Weapon From Selected Model	Wraps an imported FBX/GLTF in a socketed weapon prefab with data
7	Build Game Mode Scenes	Mode_Raid, Mode_SearchAndDestroy, Mode_Hardpoint (arena, objectives, spawns, NavMesh)
8	Build Armory	Armory hub scene (first in build settings), catalog, HDRP dynamic resolution

Command line

```
REM Full PlayMode test suite (headless)
Unity.exe -batchmode -nographics -projectPath . -runTests -testPlatform PlayMode -testResults results.xml

REM Visual check: renders PNGs of every scene (needs graphics, so no -nographics)
set FPS_CAPTURE_DIR=C:\captures
Unity.exe -batchmode -projectPath . -runTests -testPlatform PlayMode -testFilter FPS.Tests.RenderCapture

REM Windows player build (output kept outside the repository)
Unity.exe -batchmode -quit -projectPath . -buildTarget Win64 ^
  -buildWindows64Player "C:\Users\9supe\Desktop\Claude of duty Build\ClaudeOfDuty.exe"
```

> Close any editor that has this project open before running batch commands — Unity locks the project.

3 Controls

Action	Keyboard & mouse	Gamepad
Move / look	WASD / mouse	Left / right stick
Fire	Left mouse	Right trigger
Aim down sights	Right mouse	Left trigger
Sprint (hold) · tactical sprint (double-tap)	Left Shift	L3
Hold breath (while aiming)	Left Shift	L3
Walk slowly	Left Alt	—
Walking pace up / down	Mouse wheel	—
Crouch (tap) · slide when sprinting	C	B
Prone · dive when sprinting	Hold C, or Left Ctrl	Hold B
Lean left / right	Q / E	D-pad ← / →
Reload (tap) · speed reload (second tap) · magazine check (hold)	R	X
Change fire mode	B	D-pad ↓
Switch to canted sight	X	D-pad ↑
Mount weapon on ledge or corner	V	R3
Primary / secondary weapon	1 / 2	Y cycles
Interact (tap) · hold-to-use objectives (hold)	F	A
Weapon light and laser on/off	T	View
Throw lethal grenade (hold to cook)	G, mouse forward	RB
Throw tactical grenade (hold to cook)	H, mouse back	LB
Options menu · leave the workbench	Escape	Start

Tap/hold threshold is 0.35 s. The menu key lives in its own *System* input map that stays on while gameplay input is paused, so menus and stations always respond. All input goes through one `InputReader` asset: continuous state is polled, discrete intents are events.

4 Movement and the player's body

Locomotion

- **Paces** (m/s): slow walk 1.3, combat pace 3.4, sprint 5.8, tactical sprint 7.2. The mouse wheel trims walking pace in 10 % steps down to 30 %.
- **Modifiers**: crouched 55 %, prone 22 %, aiming 60 %, backwards 65 %, strafing 80 %.
- **Momentum**: acceleration 8 m/s^2, deceleration 10 m/s^2; less control while sprinting and almost none in the air.
- **Traction** comes from the ground's physics material (dynamic friction 0.6 = full grip). Ice makes you slide and overshoot.
- **Weapon load**: every kilogram carried costs 1.2 % speed (floor 82 %); ergonomics affect handling.
- **Stamina**: sprint drains 7 %/s, tactical sprint 20 %/s; regenerates 12 %/s after 1 s. Running dry exhausts you until 35 % is back.
- **Slide**: crouch while sprinting above 4.5 m/s for an 8 % boost; slope physics and friction apply, steerable, up to 2.5 s.
- **Dive**: go prone while sprinting to dive forward (0.55 s) and land prone.
- **Injuries**: leg damage cuts sprint speed by up to 45 % and blocks tactical sprint past 50 % injury.

Stance, lean and mounting

Stance	Capsule height	Eye height	Transition
Standing	1.8 m	1.66 m	stand ↔ crouch 0.28 s
Crouched	1.15 m	1.02 m	crouch ↔ prone 0.65 s
Prone	0.4 m	0.28 m	stand ↔ prone 0.9 s

- Clearance checks stop you standing up under a low roof. Prone aligns the body to slopes up to 30°, limits pitch to −30°/+25° and cannot swing through walls.
- **Lean**: ±18° about the hips, limited by sphere-casts so you never lean through a wall.
- **Mounting** (V): rest the weapon on a ledge or wall corner within reach. Sway drops to 20 % and recoil to 55 % inside a limited arc (ledges 40°×18°, corners 25°×25°).

Physiology

- **Heart rate** 65–185 bpm. Exertion drives it (sprint 142, tactical sprint 172) and suppression adds stress; rises over ~3.5 s, recovers over ~14 s.
- **Breathing** (12–42 breaths/min) drives a figure-8 weapon sway of 0.18°–1.4°, plus slow wander and a heartbeat twitch. Crouching reduces sway to 75 %, prone to 45 %; arm injuries increase it.
- **Hold breath** (Shift while aiming) steadies sway to 12 % for 2.5–7 s depending on heart rate; running out costs a gasp with heavy sway.
- **Headbob** is gait-synchronised per pace, with a landing dip proportional to fall speed, a slide lower-and-roll and a dive pitch.

Health and feedback

- 100 HP. Hit zones (head, thorax, stomach, arms, legs) follow stance, lean and prone, so enemy fire hits what is exposed.
- **Suppression**: supersonic rounds within 2 m of your head flinch it away from the crack, blur the periphery and raise heart rate.
- **Stun whiteout**: on-screen flashes blind fully, off-screen ones fade with angle; deafness muffles the mix and a tinnitus ring fades in and out.
- **Stress**: heavy damage and suppression close a vignette and drain colour. Death removes control until respawn.
- **Interaction**: look at an interactable within 2.2 m and press F (crates, locker, workbench, fallen soldiers' rifles, objectives); hold F for hold-to-use objectives with a HUD progress bar.

5 Weapon handling, aiming and recoil

The first-person weapon hangs from a chain of pivots, each written by exactly one system: **Inertia** → **Recoil** → **Action poses** → **Aim** → **weapon model**, with two-bone IK arms. Every rotation acts about the point where the stock meets the shoulder.

Inertia and physical presence

- **Spring handling**: the weapon follows the view through damped springs whose stiffness comes from mass, ergonomics and the build's measured **rotational inertia** about the shoulder. Long barrels and suppressors swing visibly slower.
- **Free aim**: quick turns lead inside an elastic deadzone (2.5°×4° at the hip, 25 % when aiming, 10 % mounted).
- **Movement**: trail and strafe roll, sprint posture, and a high-ready posture for tactical sprint.
- **Barrel collision**: a sphere-cast along the barrel pushes the weapon back near walls, then tilts it to a high ready (short barrels) or low ready (16"+). A tucked weapon cannot fire or aim.

Aiming and optics

- **Procedural ADS**: no animation. Each frame the pose is solved that places the active sight's aligner on the eye line at its eye relief. Base ADS time 0.3 s, scaled by ergonomics and parts.
- **Canted sights**: X rolls the weapon 45° onto offset irons.
- **Picture-in-picture scopes**: a second camera renders into the lens with FOV = lens angular size ÷ magnification, so the image is optically consistent. Misalignment produces an eye-relief scope shadow; the main view blurs in the periphery; look sensitivity scales with zoom.
- Red dots and irons are unmagnified and cost no extra camera.

Recoil

- Per shot: muzzle rise, biased sideways torque and roll; rearward shoulder push and a small upward kick; a delayed **bolt-carrier jolt** ~35 ms later; **permanent view climb** you pull down yourself; a damped camera concussion with no random shake.
- Ramps up during sustained fire and resets after a 0.3 s pause. Aiming 75 %, mounted 55 %. Scales with **momentum** — heavier builds kick less — and per-axis attachment multipliers.

Arms

Two-bone IK keeps the firing hand on the pistol grip. The support hand moves between the handguard, a fitted foregrip, the magazine (reloads and checks) and the charging handle (cycling).

6 Weapon mechanics and weapons

- **Data split**: `WeaponDataSO` holds the firearm (mass, ergonomics, barrel length, action, fire modes, RPM, MoA, capacity, stage timings, recoil and audio profiles); `CaliberDataSO` holds the ammunition (mass, diameter, velocity vs barrel length, drag model and BC, penetration, damage).
- **Chamber tracking**: closed-bolt and bolt-action weapons start at capacity + 1; open bolt never carries a +1. Hold-open locks the bolt back on the last round; bolt actions cycle after the trigger is released.
- **Fire modes**: semi, 3-round burst, full auto at the cyclic rate.
- **Firing**: every round leaves the **muzzle socket along the bore** with MoA dispersion — never from screen centre. Muzzle velocity comes from caliber and actual barrel length. Fire cancels sprint after a sprint-to-fire delay; tucked barrels, dives and prone transitions block it.
- **Staged reloads** (magazine out → in → charge only if empty; bolt release 0.6× when locked back): tap R = tactical (partial magazine back in the pouch); second tap during removal = speed reload (physical magazine drops); hold R = magazine check (estimate from "empty" to "full"). Sprint, dive, swap or firing the chambered round interrupts; the next press resumes after the last completed stage.
- **Swaps**: holster/draw times scale with ergonomics; each weapon keeps its rounds, chamber and reload progress while holstered.
- **Casings**: pooled physical casings eject with spin and clink per surface. **Resupply** at ammo crates; **looting** a fallen soldier's rifle takes its loaded magazines (up to loadout + 2).

Weapon	Action	Build and notes
Test Carbine 5.56	Closed bolt, semi/auto, 800 RPM, 2.5 MoA, 30 rounds, 4 spare magazines	Modular receiver with barrel, muzzle, optic, underbarrel, stock, laser and light mounts; 45° canted irons. Default build ≈ 3.47 kg: 14.5" barrel, flash hider, 4× prism scope, angled grip, fixed stock, light, laser; ≈ 899 m/s.
Test Marksman .300 WM	Bolt action, 60 RPM, 0.6 MoA, 5 rounds	6.2 kg, fixed 8× PiP scope, 24" barrel, heavy recoil, manual cycling.

7 Gunsmith and attachments

- **Slots**: barrel, muzzle, optic, underbarrel, stock, laser, light. Each part attaches at a `{Slot}Mount` anywhere under the weapon, so a barrel carries the muzzle mount.
- **Sockets move with parts**: a suppressor becomes the new muzzle (rounds and flash leave its end), a foregrip moves the support hand, a collapsed stock moves the shoulder point.
- **Live stats**: mass, ergonomics, barrel length → muzzle velocity, recoil multipliers, flash/gas size, ADS time, suppressed audio profile, and the **measured centre of mass and moment of inertia** about the shoulder (from each part's real position), which drives the handling springs.
- Builds are applied to a runtime clone of the weapon data — assets are never modified. Live refits keep rounds, chamber and reload progress.

Part	Mass	Effects
10.5" CQB barrel	0.45 kg	+6 ergonomics, flash ×1.5, ADS ×0.94, lower velocity
14.5" carbine barrel	0.60 kg	Baseline
20" marksman barrel	0.85 kg	−8 ergonomics, flash ×0.8, ADS ×1.08, full 940 m/s
Birdcage flash hider	0.08 kg	Flash ×0.25
Three-port compensator	0.12 kg	Vertical recoil ×0.8, horizontal ×0.85, flash ×1.4
5.56 suppressor	0.55 kg	No flash, suppressed sound, vertical ×0.9, rearward ×0.85, +1 % velocity, −8 ergonomics, front-heavy
4× prism scope	0.50 kg	PiP magnified optic, −4 ergonomics, ADS ×1.1
Red dot sight	0.15 kg	Unmagnified, ADS ×0.95
Vertical foregrip	0.12 kg	Horizontal recoil ×0.85, +4 ergonomics, vertical hand pose
Angled foregrip	0.08 kg	Vertical recoil ×0.92, ADS ×0.95, +6 ergonomics
Fixed stock	0.40 kg	Rearward recoil ×0.9
Collapsed stock	0.30 kg	+4 ergonomics, ADS ×0.95, rearward ×1.1, sits closer
Visible laser	0.09 kg	Beam and dot on whatever it hits
Weapon light	0.12 kg	15,000 cd spot, 40° spill; lights the volumetric fog; makes the carrier visible to AI

8 Ballistics, armour and damage

- **Simulation**: one projectile per shot with gravity and **G1/G7 drag** against the wind, sub-stepped at 1/120 s and resolved with one batched `RaycastCommand` per frame (no per-shot allocation). Up to 4 interactions per round per frame; rounds expire after 6 s or below 80 m/s.
- **Penetration**: energy cost per metre = surface resistance (0.007 × density × hardness) × bullet cross-section (mm^2) × an AP factor (1.5 → 0.5 by penetration power). Thickness is measured by casting back from the far side; exit speed comes from leftover energy, with deflection. Drywall is shot through easily; 30 cm of concrete stops rifle rounds.
- **Ricochet**: impacts shallower than the surface's ricochet angle glance off with probability 1 − angle/ricochetAngle.
- **Near misses**: supersonic rounds within 2 m of a listener produce a crack and suppression.

Calibers

Caliber	Bullet	Velocity	Pen	Dmg
5.56×45 M855	4.0 g, G7 .151	940 m/s @20"	40	35
7.62×51 M80	9.5 g, G7 .200	840 m/s @22"	55	55
9×19 124 gr	8.0 g, G1 .150	360 m/s @4"	15	30
.300 Win Mag	12.3 g, G7 .290	880 m/s @24"	70	90
Fragment	0.5 g, G1 .012	1300 m/s	12	22

5.56 loses 7.5 m/s per inch of barrel below 20".

Surfaces

Surface	Density kg/m^3	Hardness	Ricochet
Concrete	2400	60	20°
Brick	1900	45	15°
Wood	600	35	8°
Drywall	700	3	3°
Metal	7850	100	30°
Glass	2500	12	5°
Dirt	1600	35	6°
Flesh	1050	15	—

Bodies and armour

- **Health**: 100 HP. Zone multipliers: head ×3.5, thorax ×1.4, stomach ×1.1, arms ×0.55, legs ×0.7; limb capacity 40. Rounds can pass through flesh and continue.
- **Armour**: Level IV ceramic front and back plates (rating 60, durability 80, 3.2 kg each) and a Level IIIA helmet (rating 25, durability 40, 1.4 kg). Coverage depends on direction — face shots get under the helmet.
- **Defeat roll**: a logistic roll of energy-scaled penetration against protection (worn down by hits) decides stop (blunt trauma) or defeat (residual absorption).
- **Injuries**: legs slow sprinting; arms add sway.

9 Sound

- **Own shots** are layered in first person: pressure transient, mechanism and body resonance.
- **Propagation**: other shots arrive after **distance ÷ 343 m/s**; beyond the distant threshold they play as a muffled distant report.
- **Reverb tails** blend indoor, urban-canyon and open-ground tails, weighted by a 9-ray probe that measures ceiling and nearby walls around the listener.
- **Supersonic cracks** play at the point where a round passed closest to your head.
- **Muffling**: suppression, low health and stun deafness sweep a low-pass filter on the listener; flashbangs add a fading tinnitus ring.
- **Foley**: dry fire, magazine out/in, bolt, fire-mode switch and magazine check, for you and for nearby soldiers.
- **Impacts**: each surface has its own impact sound and casing clinks, delayed by distance.
- **Profiles**: suppressed rifle (quieter and duller, audible to 300 m instead of 1500 m); three explosion profiles (frag, stun, C4) with crack, boom, debris and tails.

Master-bus dynamic range (options menu)

Preset	Compressor	Makeup	Limiter	Character
War Tapes	−10 dB, 1.5:1	+4 dB	Soft, −0.5 dBFS	Loud and wide, saturating
Studio Reference	Transparent	0 dB	−0.3 dBFS ceiling	Uncoloured
Night Mode	−30 dB, 5:1	+12 dB	−6 dBFS ceiling	Lifts footsteps, tames gunfire

> **Placeholder audio.** Every clip is synthesized by the editor (`AudioSynth`) into `Art/Audio` ; soldier callouts use Windows speech synthesis in three voices. Swap in recorded clips by assigning them in the profiles, surfaces and voice assets — no code changes needed.

10 Graphics and rendering

Pipeline configuration (HDRP 17.6)

Feature	Setting
Colour space / buffer	Linear; HDR colour buffer
Lit shader mode	Deferred and forward both supported
Shadows	4096 shadow atlases, 32-bit directional shadow depth, up to 128 shadow requests
Volumetrics	On — global volumetric fog plus local fog volumes (up to 256 on screen)
Ambient occlusion	SSAO on
Reflections / GI	SSR and SSGI off; ray tracing off
Decals	On (bullet holes)
Distortion	On (barrel heat shimmer)
Motion vectors	On (motion blur, TAA)
Lens flare	Screen-space lens flare supported
Dynamic resolution	Hardware scaling 50–100 %, TAA upscale, DLSS and FSR2 when the platform reports them

Cameras and post-processing

- **Main camera** renders the world and the first-person weapon in-world (no separate overlay camera), so the weapon receives real lighting and shadows.
- **PiP scope camera** (one per player) renders to an `R16G16B16A16_SFloat` target fed to the HDRP/Unlit lens through a property block. Volumes are split by layer: a scope-only volume adds the eye-relief vignette, turns tonemapping off (the main camera tonemaps the lens once) and disables motion blur; a main-only volume adds peripheral depth of field while scoped.
- **Feedback volumes** on the main camera: suppression focal blur (DoF), stress vignette and desaturation, stun whiteout, and the motion-blur setting.

Lighting and atmosphere

- Thin global volumetric fog (mean free path 250 m) so flashlight cones and muzzle flashes read in the air; dust volumes in interiors make light shafts visible.
- Interior lamps are physically based point lights in lumens (workshop 1,800 lm, range 2,200 lm, soft shadows); weapon lights are spot lights in candela that light the fog.
- AI sight uses the same light: sun shadow ray, open sky versus roofs and walls, and flashlight cones.

Effects

- **Muzzle flash**: a one-frame HDRP point light (40,000 lm) that lights dark rooms, scaled by the build's flash multiplier (0 when suppressed), plus a gas burst per shot.
- **Barrel heat** (0–1, exponential cooling) drives heat-haze distortion and lingering breech smoke.
- **Impacts**: pooled HDRP decal projectors sized by caliber, parented to the hit collider, plus dust puffs tinted per surface. Hits on your own body show nothing in your view.
- **Particles** use HDRP/Unlit with a 4×4 alpha flipbook (HDRP/Unlit has no vertex colour); tint per instance through a property block.
- **Smoke grenades** are a local volumetric fog volume plus billowing particles.

HUD

Minimal and built in code (uGUI): compass tape with bearing, interaction prompts, fire-mode and magazine status that appears only on a mode switch or magazine check, stress vignette and desaturation, and no crosshair by default. Match HUD: mode, objective, clock, score, round banners and a hold-progress bar.

11 Throwables and equipment

- **Throwing**: press lowers the rifle and pulls the pin (0.35 s); the spoon flies and the fuse burns while held (**cooking**). Release throws along the view with 10° loft plus your own velocity, keeping the fuse already used. Hold too long and it goes off in hand; dying or being disabled mid-cook drops it live at your feet.
- **Default loadout**: 2 frags (lethal) and 2 stun grenades (tactical). The equipment locker swaps stun ↔ smoke; ammo crates restock grenades.

Item	Fuse	Effect
M67 frag	4.5 s	0.4 kg. **64 real fragment projectiles** through the ballistics sim (they penetrate, ricochet and suppress); blast overpressure within 4 m, blocked by solid cover; concussive push on casings, magazines, rifles and bodies.
M84 stun	1.5 s	Blinds fully within 4 m, fading to 18 m, by facing and line of sight; deafens to 25 m. Through a wall: 15 % blindness, 60 % deafness.
M18 smoke	1.5 s	≈ 4.5 m cloud: builds 3 s, holds 35 s, fades 6 s. Local volumetric fog + particles, and an obstruction AI cannot see through (thin edges partly).
C4 charge	40 s	Search & Destroy objective: 10 m blast, 48 fragments, large report, quickening beep.

12 Enemy AI

The soldier

- Procedural body on a NavMesh agent: crouch/stand heights, ±22° lean from the hips, body turns to follow the aim, shoulder aim pivot with turn-rate limit and recoil, head look, IK arms on the rifle.
- Carries the same gunsmith-built carbine as the player (red dot, light, flash hider) with identical ballistics and audio; rounds leave its muzzle along its bore; magazines and chamber are counted; reloads use real timings and foley.
- Kit: one frag, one stun grenade, a helmet, one of three voices. On death the body collapses physically and drops a lootable rifle.

Perception (≈ 10 Hz)

- **Sight**: awareness builds per target from visible body fraction (head, chest, pelvis lines), **light level** (dark corners shrink sight range to 12 %), stance (crouched 65 %, prone 35 %), motion, distance (90 m), and focus vs peripheral vision (30° / 80°, peripheral at 35 % speed). Under 4 m it is three times faster. Full awareness = identified ("Contact!"), remembered for 25 s.
- **Smoke** thins or blocks sight lines; muzzle flashes still glow through thin smoke and give an unsuppressed shooter away instantly.
- **Hearing** alerts but never identifies: footsteps by speed and stance (halved through walls); gunshots to the weapon's audible range, with suppressed shots located far worse (30 % vs 8 % error).
- **Stress**: near misses and hits raise suppression and turn the soldier toward the shooter; stun grenades blind and deafen.

Tactics

- **Idle / suspicious**: hold a post or patrol; look toward noises. **Alerted**: move up on the last known position, pre-aiming around corners (slicing the pie).
- **Firefight**: NavMesh cover search against the threat — low cover (crouch to hide, stand to fire) or tall cover (lean out) — avoiding teammates' claims; peek-and-hide cycles; 0.35–0.7 s reaction; aim error grows with own suppression, target speed and moving fire, and settles while tracking; bursts of 3–6; never fires into its own cover or through a teammate.
- **Lost contact**: suppressive fire on the last known position for 6 s, then hunt. A target holding still behind cover for 4 s gets a **cooked frag** (6–28 m, arc solved to clear the cover).
- **Reloading**: callout, duck into cover, a teammate covers.

Squads

- **Shared knowledge**: one sighting engages the whole squad.
- **Suppress and flank**: when the target goes to ground, one soldier flanks ("Flanking left/right!") while the rest suppress; "Target suppressed!" when squad rounds crack past the target's head.
- **CQB room clearing**: stack on the nearest door, flashbang in while the stack looks away, then one man at a time slices the pie through the doorway to his point of domination; "Clear!" if empty.
- **Coordination**: grenade cooldowns, cover claims, team-wide friendly-fire checks (player included), "Covering!", "Man down!".
- **Objectives**: hold a point (spread around it) or work it (plant/defuse while others cover); a dead worker is replaced; threats are not chased beyond a 25 m leash.
- **Callouts**: Contact, Reloading, Covering, Flanking left/right, Target suppressed, Frag out, Flashbang out, Stack up, Go go go, Clear, Lost contact, Man down.

13 Game modes

All modes run on a **match director**: the player plus AI soldiers per team in squads (friendlies tan, hostiles olive); phases pre-round countdown → live → round over → match over; clock, round wins and scores; respawns after the mode's delay (or one life) with the player's controls removed while dead; fresh soldier bodies each spawn and corpse clean-up.

Smart spawns score every point: −60 if a living enemy can see a body standing there; distance to the nearest enemy (capped at 40 m) and −100 inside 12 m; +4 per nearby teammate (up to 3); a penalty for points used in the last 5 s.

Mode	Rules	Teams
Tactical Raid / Extraction	One life, no resupply. Complete the objectives by holding Interact (download intel 6 s, plant tracker 5 s); each completion raises the alarm across the garrison. Then reach the LZ and hold inside it for 8 s. Death or the 15-minute clock fails the mission. Ammunition is what you carry plus magazines looted from the dead.	Solo vs a 6-man garrison in two squads
Search & Destroy	120 s rounds, one life. Attackers plant C4 at site A (warehouse) or B (container yard): 5 s hold, 40 s fuse with a quickening beep. Defenders stop the plant or defuse (7 s hold). Elimination wins unless a planted charge still needs defusing; a planted charge stops the clock. Sides swap after round 3; first to 4 wins. AI attackers send a planter and cover him; defenders split over the sites; everyone converges on a planted charge.	Player + 2 vs 3
Hardpoint	One of three zones is live for 60 s, then moves. Each second a team alone in the zone scores; contested zones score nobody. First to 150, or most after 10 minutes. 4 s respawns away from enemy sight and outside the live zone; all squads are sent to the live zone.	Player + 3 vs 4

14 Armory, profiles and settings

The Armory hub

- A lit workshop with the **gunsmith workbench**, an ammo crate and the equipment locker, opening straight onto a **30 m indoor range** (targets at 10, 20 and 28 m; an unarmoured dummy at 15 m and an armoured one at 25 m). Test-fire a build seconds after making it.

Gunsmith workbench

- Interact to step up: the bench camera shows the carried weapon on a turntable with a marker on every mount.
- Drag to turn it; click a marker to list compatible parts from the catalog; click a part to fit it. The model, measured balance and stats panel (mass, ergonomics, barrel, muzzle velocity, recoil, ADS time, swing inertia, flash or suppressed) update instantly.
- "Fit to weapon" applies the build to your rifle and saves it; "Leave" or Escape hands control back.

Profile and saves

- `profile.json` in the persistent data folder: weapon blueprints (weapon and parts by asset id), loadout (a blueprint per slot plus grenade ids) and all settings.
- Atomic writes (`.tmp` then swap) keep a `.bak`; an unreadable file is moved to `.corrupt` and the backup or defaults load. The saved loadout is applied on every spawn.
- Catalog ids are asset names — renaming an asset orphans it in saved profiles (missing parts are skipped with a warning).

Options menu (Escape) — applied live, saved with the profile

Setting	Values
Field of view	60–100° in 5° steps
Scope resolution	512 / 1024 / 2048 px PiP target
Motion blur	Off / Low / High
Upscaling	Native, TAAU, FSR2, DLSS (FSR2 and DLSS only where detected, otherwise TAAU)
Render scale	50 / 59 / 67 / 77 / 87 / 100 %
Audio dynamic range	War Tapes / Studio Reference / Night Mode

The menu pauses the player while open; at the workbench, Escape leaves the bench instead.

15 Maps and scenes

Scene	Contents
Armory hub, first in build	Workshop, gunsmith workbench, ammo crate, equipment locker, 30 m indoor range with targets and armoured dummies.
Locomotion Test Range sandbox	Movement: 15° and 35° slide ramps, ice sheet, standing and crouched mounting ledges, corner wall, crouch (1.3 m) and crawl (0.6 m) tunnels, narrow barrel-tuck corridor. Shooting: range lane with targets at 25/50/100/200 m; penetration wall row (drywall, wood door, glass, 1 cm and 3 cm steel, brick, concrete) with plain and armoured dummies. CQB and acoustics: dusty dark shoot house (a CQB room), street canyon. Ammo crate, equipment locker, and a cover yard held by Squad Alpha (facing away until gunfire brings them).
Mode_Raid · **Mode_SearchAndDestroy** · **Mode_Hardpoint**	One arena: roofed two-room warehouse (both rooms CQB-clearable), walled compound to the west, container yard to the east, cover lines across the middle; each scene carries its mode's director, objectives, spawns and HUD.

Every scene has a baked NavMesh, volumetric atmosphere and the full player prefab.

16 Architecture

Assemblies (references only point down the table)

Assembly	Namespace	Scope
FPS.Core	FPS.Core	Input, EventBus, pooling, layers, spring and IK math, shared interfaces and event structs
FPS.Ballistics	FPS.Ballistics	Calibers, projectile sim, penetration/ricochet, casings, hitboxes, health and armour
FPS.Audio	FPS.Audio	Acoustics, weapon audio profiles, impact decals/dust/audio, particles, dynamic range
FPS.Player	FPS.Locomotion	Locomotion, stance/lean/mount, breathing, suppression and flash feedback, camera chain
FPS.Weapons	FPS.Weapons	Weapon data, model rig and sockets, viewmodel, optics, recoil, controller, gunsmith, equipment
FPS.AI	FPS.AI	Perception, soldier body and weapon, tactical brain, squads, CQB rooms
FPS.UI	FPS.UI	Diegetic HUD
FPS.Gameplay	FPS.Gameplay	Match director, modes, objectives, spawns, match HUD, profiles, workbench, settings
FPS.Editor	FPS.EditorTools	Builders, wizards, validators (editor only)

Upward communication uses the zero-allocation `EventBus<T>` (struct events: ShotFired, Gunshot, WeaponAction, Suppression, ProjectileImpact, Damage, Flashbang, SquadCallout, MatchPhase), interfaces declared in Core, or push-style setters.

Player rig — one writer per transform

```
Player            CharacterController, PlayerRig, Locomotion (yaw), StanceAndLean, HeadbobAndBreathing
└ BodyPivot       StanceAndLean: torso height, prone ground pitch
  └ LeanPivot     StanceAndLean: lean roll (±18°)
    └ HeadPivot   StanceAndLean: eye offset (stance + mount height)
      └ PitchPivot    Locomotion: look pitch
        └ HeadMotion  Headbob/Breathing: bob, landing, slide/dive posture, camera share of sway
          ├ CameraOffset      PhysicalRecoilEngine: damped camera concussion
          │ └ PlayerCamera    ADS controller writes field of view
          └ ViewmodelRoot     layer FirstPersonViewmodel
            ├ InertiaPivot    springs, free aim, trail, postures, sway, barrel tuck
            │ └ RecoilPivot   per-shot kick about the stock
            │   └ ActionPivot reload, magazine check, bolt, holster/draw
            │     └ AdsPivot  hip ⇔ solved sight alignment → weapon model
            └ Arms            two-bone IK to the weapon's grip sockets
```

Layers

Layer	#	Use
Player	8	Player and AI capsules
FirstPersonViewmodel	9	Weapon/arms; collides with nothing
WorldGeometry	10	Level geometry; ballistics, tuck, lean
Hitbox	11	Per-bone damage colliders
Interactable	12	Interaction targets
Debris	13	Casings, magazines, grenades

Engineering rules

- Zero allocation on hot paths: pools for casings, magazines, decals, emitters, grenades; non-allocating physics; batched bullet raycasts.
- Domain-reload safe: statics reset at SubsystemRegistration.
- SI units everywhere; imperial only as authoring fields.
- Data-driven ScriptableObjects (`*SO`); models bind through sockets, never hardcoded paths.
- Bullets always spawn from the `MuzzleSocket` along the bore.
- No stubs or TODOs anywhere.

The repository's `CLAUDE.md` is the authoritative engineering guide and lists every contract between systems.

17 Content pipeline and tests

Generated content

- **Weapons and parts**: built from primitives by builder 3, with sockets, data assets, recoil profiles and audio profiles.
- **Importing a real weapon**: select an FBX/GLTF → builder 6. It wraps the model under a `WeaponModelRig`, converts materials to HDRP/Lit, strips colliders, guesses sockets and magazine renderers and creates `<Name>_Data`. The rig inspector lists validation issues with fix buttons; place sockets with colour-coded gizmos and add a `WeaponSight` to optics.
- **Soldiers, grenades, voices, scenes, NavMeshes**: builders 5, 7 and 8.
- **Audio**: `AudioSynth` writes every clip; voices come from Windows TTS.

Tests (PlayMode, NUnit + InputTestFixture)

Suite	Covers
Core, Locomotion	Event bus, pooling, springs, IK, paces, stances, lean, slide, stamina
Weapon, WeaponController	Recoil, ADS, PiP, chamber tracking, fire modes, staged and interrupted reloads, magazine checks, swaps
Ballistics, Effects	Drag, penetration, ricochet, armour, near misses, acoustics, decals, muzzle VFX
Gunsmith	Stat maths, measured inertia, socket moves, live refits, devices
AIPerception, AITactics	Sight, light, hearing, smoke; cover, suppression, flanking, room clearing, frag flushing
Equipment, Gameplay	Cooking, frags, stun, smoke; all three modes, spawns, objectives
Armory	Profiles (atomic save, corruption recovery), workbench, settings, looting
RenderCapture (opt-in)	PNG captures of the range, modes and armory

Latest full run: **89 total, 84 passed, 0 failed, 5 skipped**. New non-trivial logic ships with a test in `Assets/_Project/Tests`.

> **Art status.** All geometry is built from primitives with HDRP/Lit materials; there are no character or weapon models and no animation clips (bodies, arms and weapon motion are procedural). Builder 6 turns an imported FBX/GLTF into a socketed weapon prefab.

18 Known issues and workarounds

Issue	Detail and handling
No scene flow in the build	The player boots into the Armory with no way to reach the mode scenes, and the options menu has no Quit button (exit with Alt+F4). In the editor, open a mode scene directly. A follow-up task to add mode selection, "Return to Armory" and Quit has been proposed.
HDRP 17.6 fog bug (editor only)	Entering Play mode resets HDRP's local volumetric fog manager after the pipeline allocated its buffers, so every frame threw "Render Graph Execution error" in the fog voxelization pass. Fixed by `VolumetricFogBufferRepair` (commit `1b05b0a`), which re-allocates the buffers before the first scene loads. Built players are unaffected. Remove it once HDRP ships a fix.
Editor input needs Game view focus	Keyboard and mouse only reach the game after clicking inside the Game view (Input System default). Simulated-input tests confirm movement and firing work in every scene.
Harmless editor log noise	An `ArgumentOutOfRangeException` from Unity Search indexing appears once at editor start-up; it is Unity-internal.
Placeholder art and audio	Primitives, procedural animation, synthesized sound and TTS voices. Replace by importing models (builder 6) and assigning recorded clips — no code changes.
Workbench scope	Only weapons with attachment mounts offer parts; the marksman rifle's scope and barrel are fixed.
Upscalers	DLSS and FSR2 depend on hardware/platform support and fall back to TAAU.
Single-player	No networking; AI difficulty is only its tunable component values.
Housekeeping	Build output lives outside the repo (`Desktop\Claude of duty Build`). A second, stale clone exists at `Desktop\Mage Match\Claude-of-duty` (Milestone 1 only) — don't work there. `GAME_OVERVIEW.md` and `ProjectSettings/PackageManagerSettings.asset` are uncommitted.

19 Recommended next steps

1. **Front end and scene flow**: main menu or in-Armory deployment board for the three modes, "Return to Armory", Quit, and a loading screen.
2. **Art pass**: import real weapon models through builder 6; character models with retargeted procedural IK; environment kits replacing primitive blockouts.
3. **Audio pass**: recorded gunshots (close/distant/suppressed), mechanism foley, impacts, explosions and voice lines into the existing profiles.
4. **Content**: more weapons and calibers (the 7.62 and 9 mm data already exist without a weapon), more parts, maps built for each mode.
5. **AI tuning**: difficulty presets over perception and aim parameters; more voices.
6. **Settings**: key rebinding, mouse sensitivity and audio volume sliders.
7. **Release hygiene**: remove the HDRP fog workaround when Unity fixes the bug; add a build menu item and CI for the test suite.